

Reviewer Pack — Coxeter Matrix Spectral Gap and Resolution Proof Width

Entry a0adfbf5b523 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Coxeter Matrix Spectral Gap and Resolution Proof Width

Entry ID: a0adfbf5b523 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-29 13:16:11 UTC

1. Conjecture

Field A (mathematical branch): Combinatorial Geometry (Coxeter Groups) **Field B** (complexity object): Resolution proof complexity

Statement:

For any given instance of a SAT problem with n variables, the spectral gap γ of its associated Coxeter matrix W is upper-bounded by the resolution proof width $w(W)$ for that instance, i.e., $\gamma \leq C * w(W)$, where C is an absolute constant.

Rationale (proposer's reasoning):

Coxeter groups have been previously used to study structural properties of graphs and matrices. By leveraging the spectral gap property of Coxeter matrices, we can potentially relate the complexity of solving SAT problems to their underlying graph structure, which is captured by the resolution proof width.

Taxonomy category: c003b_cumulative_entropy (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): bb92361464ff87a0

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all generated 3-CNF formulas with n variables ($n \leq 40$), the spectral gap γ of the Coxeter matrix W is less than or equal to C times the resolution proof width $w(W)$ for each formula, where $\gamma \leq C * w(W)$. The criterion is falsified if there exists any formula where $\gamma > C * w(W)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Coxeter matrix spectral gap" AND "resolution proof complexity" - "SAT problem" AND "Coxeter group" AND "proof width" - "Combinatorial geometry" AND "resolution proof complexity" AND "spectral gap"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.5s

5.1 Generated Python source

```
#!/usr/bin/env python3
"""C-003b cumulative entropy – FAITHFUL reference harness.

Computes the REAL cumulative proof entropy
    Phi(pi) = sum_t |activeClauses(sigma_t)|
over an actual resolution refutation (Davis-Putnam variable elimination)
of Tseitin formulas, exactly per the Lean definitions in Conjecture003.lean:
    proofStateEntropy(sigma) := sigma.activeClauses.card
    cumulativeEntropy(states) := sum (map proofStateEntropy states)
plus the width-aware totalLiteralWeight version (sum of clause sizes).

This is NOT the CDCL clause-count proxy used by the old c003b_counterexample.py:
it tracks the active clause database at EVERY elimination step of a genuine
resolution derivation, so it is the actual Phi the formalization names.

Pure stdlib (no networkx/pysat), so it runs in the pipeline sandbox.
Protocol: run_trial(seed)->dict with TRIAL/RESULT lines.

Author: Ludovico Kubler (harness by the SEC engine, hand-verified).
NOTE: no `from __future__ import annotations` – the sandbox injects a prelude
above the file, which would break that statement's must-be-first rule.
"""

import json
import math
import random
import sys

Clause = frozenset # a clause = frozenset of signed ints (var or -var)

# ----- Tseitin formula generation -----

def random_regular_graph(n: int, degree: int, rng: random.Random):
```

```

"""Simple random regular graph via configuration model with retries."""
if (n * degree) % 2 != 0:
    n += 1
for _ in range(200):
    stubs = []
    for v in range(n):
        stubs += [v] * degree
    rng.shuffle(stubs)
    edges = set()
    ok = True
    for i in range(0, len(stubs), 2):
        u, w = stubs[i], stubs[i + 1]
        if u == w or (min(u, w), max(u, w)) in edges:
            ok = False
            break
        edges.add((min(u, w), max(u, w)))
    if ok and len(edges) == n * degree // 2:
        return n, sorted(edges)
# fallback: cycle + chords
edges = {(i, (i + 1) % n) for i in range(n)}
return n, sorted((min(a, b), max(a, b)) for a, b in edges)

def tseitin_cnf(n: int, edges: list, charge: dict) -> list:
    """Tseitin CNF over edge variables (1-indexed). For each vertex, clauses
    enforce parity of incident edges = charge[v]."""
    evar = {}
    for i, (u, v) in enumerate(edges):
        evar[(u, v)] = i + 1
        evar[(v, u)] = i + 1
    inc = {v: [] for v in range(n)}
    for (u, v) in edges:
        inc[u].append(evar[(u, v)])
        inc[v].append(evar[(u, v)])
    clauses = []
    for v in range(n):
        vars_ = inc[v]
        k = len(vars_)
        tgt = charge.get(v, 0)
        for mask in range(1 <= k):
            if bin(mask).count("1") % 2 != tgt:
                cl = []
                for j, var in enumerate(vars_):
                    cl.append(-var if (mask >> j) & 1 else var)
                clauses.append(frozenset(cl))
    return clauses

# ----- Resolution by Davis-Putnam variable elimination -----

def resolve(c1: frozenset, c2: frozenset, var: int):
    """Resolvent of c1,c2 on var (c1 has +var, c2 has -var). None if tautology."""
    r = (c1 - {var}) | (c2 - {-var})
    for lit in r:
        if -lit in r:
            return None # tautology

```

```
return frozenset(r)
```

```
def dp_refutation_phi(clauses: list, rng: random.Random):  
    """Run DP elimination in min-occurrence order; track the active clause DB  
    after each elimination. Returns (phi_count, phi_weight, n_steps, derived_empty).
```

```
    phi_count = sum_t |DB_t| (proofStateEntropy = card)  
    phi_weight = sum_t sum_{C in DB_t} |C| (totalLiteralWeight version)  
    """
```

```
    db = set(clauses)  
    variables = set(abs(l) for c in db for l in c)  
    phi_count = len(db)  
    phi_weight = sum(len(c) for c in db)  
    n_steps = 1  
    derived_empty = frozenset() in db  
    MAX_DB = 200000 # guard against blow-up on bad instances
```

```
    while variables and not derived_empty:  
        # min-occurrence elimination order (standard good heuristic)  
        occ = {v: 0 for v in variables}  
        for c in db:  
            for l in c:  
                if abs(l) in occ:  
                    occ[abs(l)] += 1  
        var = min(variables, key=lambda v: occ[v])  
        variables.discard(var)  
  
        pos = [c for c in db if var in c]  
        neg = [c for c in db if -var in c]  
        others = [c for c in db if var not in c and -var not in c]  
  
        resolvents = set()  
        for cp in pos:  
            for cn in neg:  
                r = resolve(cp, cn, var)  
                if r is not None:  
                    resolvents.add(r)  
                    if len(r) == 0:  
                        derived_empty = True  
        new_db = set(others) | resolvents  
        # forward subsumption (keep minimal clauses) – cheap pass  
        db = new_db  
        phi_count += len(db)  
        phi_weight += sum(len(c) for c in db)  
        n_steps += 1  
        if len(db) > MAX_DB:  
            break
```

```
    return phi_count, phi_weight, n_steps, derived_empty
```

```
# ----- Separator (cheap balanced-ish) -----
```

```
def approx_separator_size(n: int, edges: list) -> int:  
    """Cheap proxy for balanced separator size: min vertices whose removal
```

```

splits the graph ~in half. We use a simple BFS-layer cut."
adj = {v: set() for v in range(n)}
for (u, v) in edges:
    adj[u].add(v); adj[v].add(u)
# BFS from 0, cut at the layer crossing the median
from collections import deque
seen = {0}; layer = {0: 0}; q = deque([0])
while q:
    x = q.popleft()
    for y in adj[x]:
        if y not in seen:
            seen.add(y); layer[y] = layer[x] + 1; q.append(y)
if len(seen) < n: # disconnected
    return 0
half = n // 2
# vertices on the boundary between the first ~half and the rest
order = sorted(range(n), key=lambda v: layer.get(v, 0))
side_a = set(order[:half])
cut = set()
for (u, v) in edges:
    if (u in side_a) != (v in side_a):
        cut.add(u if u not in side_a else v)
return len(cut)

```

----- Trial protocol -----

```

def run_trial(seed: int) -> dict:
    rng = random.Random(seed)
    # sweep several small sizes; report the scaling exponent of Phi vs n
    ns = [6, 8, 10, 12, 14]
    data = []
    for n0 in ns:
        n, edges = random_regular_graph(n0, 3, rng)
        charge = {v: 0 for v in range(n)}
        charge[0] = 1 # odd total -> UNSAT
        clauses = tseitin_cnf(n, edges, charge)
        phi_c, phi_w, steps, empty = dp_refutation_phi(clauses, rng)
        sep = approx_separator_size(n, edges)
        data.append({"n": n, "m_edges": len(edges), "sep": sep,
                    "phi_count": phi_c, "phi_weight": phi_w,
                    "steps": steps, "derived_empty": empty})
    # scaling: log-log slope of phi_count vs n
    xs = [math.log(d["n"]) for d in data if d["phi_count"] > 0]
    ys = [math.log(d["phi_count"]) for d in data if d["phi_count"] > 0]
    if len(xs) >= 2:
        mx = sum(xs) / len(xs); my = sum(ys) / len(ys)
        num = sum((x - mx) * (y - my) for x, y in zip(xs, ys))
        den = sum((x - mx) ** 2 for x in xs)
        slope = num / den if den else 0.0
    else:
        slope = 0.0
    biggest = data[-1]
    # Sub-conjecture under test (SC1): Phi >= sep^2 on the largest instance
    # (a concrete, falsifiable scaling claim derived from the C-003b thesis
    # that cumulative entropy is forced by the separator).

```

```

holds = biggest["phi_count"] >= max(1, biggest["sep"]) ** 2
return {
    "metric_name": "phi_count_loglog_slope_vs_n",
    "metric_value": round(slope, 4),
    "instances_tested": len(ns),
    "n_max": max(ns),
    "conjecture_holds": holds,
    "counterexample": "" if holds else
        f"n={biggest['n']} sep={biggest['sep']} phi={biggest['phi_count']} < sep^2",
    "detail": data,
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [11, 23, 37]
    results = []
    for s in seeds:
        r = run_trial(s)
        results.append(r)
        print("TRIAL: " + json.dumps(r))
    slopes = [r["metric_value"] for r in results]
    holds = sum(1 for r in results if r["conjecture_holds"])
    mean = sum(slopes) / len(slopes)
    if holds == len(results):
        print(f"RESULT: SUPPORTED mean_slope={mean:.4f} support_fraction=1.0")
    elif holds == 0:
        print(f"RESULT: FALSIFIED counterexample=\"phi < sep^2\" first_failing_seed={seeds[0]}")
    else:
        print(f"RESULT: INCONCLUSIVE mixed holds={holds}/{len(results)} mean_slope={mean:.4f}")

```

6. Per-seed results

Seed	Metric value	Holds?	Counterexample
?	2.4348	☐	
?	2.0177	☐	
?	2.0528	☐	
?	2.4751	☐	
?	2.4691	☐	
?	2.2173	☐	
?	2.2983	☐	
?	2.7629	☐	
?	2.3782	☐	
?	2.7571	☐	
?	2.6234	☐	
?	2.5729	☐	
?	2.3297	☐	

Aggregate statistics:

Statistic	Value
n_seeds	13
metric_mean	2.4145615384615384
metric_std	0.23518233897149377

Statistic	Value
metric_ci95_half	0.13045568957619594
metric_min	2.0177
metric_max	2.7629
support_fraction	1.0

7. Test stdout (last 2KB)

```
"m_edges": 15, "sep": 3, "phi_count": 591, "phi_weight": 2826, "steps": 16, "derived_empty": true}, {"
TRIAL: {"metric_name": "phi_count_loglog_slope_vs_n", "metric_value": 2.2983, "instances_tested": 5,
TRIAL: {"metric_name": "phi_count_loglog_slope_vs_n", "metric_value": 2.7629, "instances_tested": 5,
TRIAL: {"metric_name": "phi_count_loglog_slope_vs_n", "metric_value": 2.3782, "instances_tested": 5,
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test code focuses on computing the cumulative proof entropy for resolution refutations of Tseitin formulas, which is a different metric than the spectral gap γ of the associated Coxeter matrix W . The conjecture states that γ is upper-bounded by the resolution proof width $w(W)$, but the test does not compute or measure the spectral gap directly.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code focuses on computing cumulative proof entropy for resolution refutations, which is a different metric than the spectral gap γ of the associated Coxeter matrix W . The critic challenged the validity of the test with respect to the conjecture, and the pre-registered support condition was not unambiguously met. | next: Re-evaluate the test code to ensure it measures the spectral gap γ directly and compare it with the resolution proof width $w(W)$ as stated in the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 7

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	20491
2	propose	ollama_remote	glm4:latest	0	0	12586
3	preregistration	ollama_remote	glm4:latest	0	0	9415
4	novelty	ollama_remote	glm4:latest	0	0	8358
5	novelty	ollama_remote	glm4:latest	0	0	14850
6	critic	ollama_remote	glm4:latest	0	0	17511
7	judge	ollama_remote	glm4:latest	0	0	9581

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 92792 ms total latency. Provider mix: {'ollama_remote': 7}

(full prompt+response transcripts available in [research/audit/a0adfbf5b523.jsonl](#))

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/a0adfbf5b523.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/a0adfbf5b523.tar.gz` (if generated)